

CHAPTER III.3-4, WEEK 11

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- 3.1. Let X be a noetherian scheme. Show that X is affine if and only if X_{red} is affine.
 [Hint: Use (3.7), and for any coherent sheaf $\mathcal{F}$ on X , consider the filtration $\mathcal{F} \supset \mathcal{N} \cdot \mathcal{F} \supset \mathcal{N}^2 \cdot \mathcal{F} \supset \cdots$, where $\mathcal{N}$ is the sheaf of nilpotent elements on X .]

Proof. If $X = \text{Spec } A$ is affine, then $X_{\text{red}} = \text{Spec}(A/\text{nil } A)$ is affine. Conversely, suppose that X_{red} is affine. For any quasi-coherent sheaf $\mathcal{F}$ on X , consider the filtration $\mathcal{F} \supset \mathcal{N}\mathcal{F} \supset \mathcal{N}^2\mathcal{F} \supset \cdots$, where $\mathcal{N}$ is the sheaf of nilpotent elements on X .

Noetherianity of X gives us a finite affine open cover $\{U_i\}$, where each $\mathcal{N}(U_i)$ is a finitely generated $\mathcal{O}_X(U_i)$ -module since $\mathcal{O}_X(U_i)$ is noetherian. Thus, we can take $n \geq 0$ such that $\mathcal{N}^n(U_i) = 0$ for each i , whence $\mathcal{N}^n = 0$. This gives us a finite filtration

$$0 = \mathcal{N}^n \mathcal{F} \subset \cdots \subset \mathcal{N} \mathcal{F} \subset \mathcal{F},$$

and thus the short exact sequences

$$(1) \quad 0 \rightarrow \mathcal{N}^{k+1} \mathcal{F} \rightarrow \mathcal{N}^k \mathcal{F} \rightarrow \frac{\mathcal{N}^k \mathcal{F}}{\mathcal{N}^{k+1} \mathcal{F}} \rightarrow 0$$

for all $0 \leq k < n$.

Clearly

$$\mathcal{N} \subset \text{Ann} \left(\frac{\mathcal{N}^k \mathcal{F}}{\mathcal{N}^{k+1} \mathcal{F}} \right),$$

so there is a quasi-coherent scheme $\mathcal{G}$ on X_{red} such that

$$\frac{\mathcal{N}^k \mathcal{F}}{\mathcal{N}^{k+1} \mathcal{F}} \cong i_* \mathcal{G}$$

with $i : X_{\text{red}} \rightarrow X$ the inclusion. It follows from (2.10) that

$$H^\bullet(X, \frac{\mathcal{N}^k \mathcal{F}}{\mathcal{N}^{k+1} \mathcal{F}}) \cong H^\bullet(X_{\text{red}}, \mathcal{G}).$$

(3.7) implies $H^i(X_{\text{red}}, \mathcal{G}) = 0$ for $i > 0$, so the long exact sequence of homology for (1) gives us exactness of

$$H^i(X, \mathcal{N}^{k+1} \mathcal{F}) \rightarrow H^i(X, \mathcal{N}^k \mathcal{F}) \rightarrow H^i(X_{\text{red}}, \mathcal{G}) = 0$$

for all $i > 0$. $H^\bullet(X, \mathcal{N}^n \mathcal{F}) = 0$, so induction gives us $H^i(X, \mathcal{N}^k \mathcal{F}) = 0$ for all $i > 0$ and $0 \leq k \leq n$. In particular, $H^i(X, \mathcal{F}) = 0$ for all $i > 0$. $\mathcal{F}$ is an arbitrary quasi-coherent sheaf on X , so that X is affine by (3.7). $\square$

- 3.2. Let X be a reduced noetherian scheme. Show that X is affine if and only if each irreducible component is affine.

Proof. One implication is clear. Suppose, then, that each irreducible component of X is affine. X is noetherian, so we can write a finite irreducible decomposition $X = Y_1 \cup \cdots \cup Y_r$, with each $Y_1, \dots, Y_r$ affine. By (Ex. 3.1), we may assume X is reduced. Now, fix a quasi-coherent sheaf $\mathcal{F}$ on X . We have a filtration

$$\mathcal{F} \supset \mathcal{I}_{Y_1} \mathcal{F} \supset \mathcal{I}_{Y_1 \cup Y_2} \mathcal{F} \supset \cdots \supset \mathcal{I}_{Y_1 \cup \cdots \cup Y_r} \mathcal{F} = \mathcal{I}_X = 0,$$

where each $Y_1 \cup \cdots \cup Y_k$ is given the reduced induced subscheme structure. This gives us exact sequences

$$0 \rightarrow \mathcal{I}_{Y_1 \cup \cdots \cup Y_{k+1}} \mathcal{F} \rightarrow \mathcal{I}_{Y_1 \cup \cdots \cup Y_k} \mathcal{F} \rightarrow \frac{\mathcal{I}_{Y_1 \cup \cdots \cup Y_k} \mathcal{F}}{\mathcal{I}_{Y_1 \cup \cdots \cup Y_{k+1}} \mathcal{F}} \rightarrow 0$$

for all $0 \leq k < r$, where

$$\mathcal{I}_{Y_{k+1}} \subset \text{Ann} \left(\frac{\mathcal{I}_{Y_1 \cup \cdots \cup Y_k} \mathcal{F}}{\mathcal{I}_{Y_1 \cup \cdots \cup Y_{k+1}} \mathcal{F}} \right).$$

Then by (2.10) we have quasi-coherent sheaves $\mathcal{G}_i$ on Y_i such that

$$H^\bullet(X, \frac{\mathcal{I}_{Y_1 \cup \cdots \cup Y_k} \mathcal{F}}{\mathcal{I}_{Y_1 \cup \cdots \cup Y_{k+1}} \mathcal{F}}) \cong H^\bullet(Y_i, \mathcal{G}_i).$$

each Y_i is affine, so $H^p(Y_i, \mathcal{G}_i) = 0$ for all $p > 0$. Hence, the maps $H^p(X, \mathcal{I}_{Y_1 \cup \cdots \cup Y_{k+1}} \mathcal{F}) \rightarrow H^p(X, \mathcal{I}_{Y_1 \cup \cdots \cup Y_k} \mathcal{F})$ are surjective for all $0 \leq k < r$ and $p > 0$ as in (Ex. 3.1). It follows that $H^p(X, \mathcal{F}) = 0$ for all $p > 0$ by induction on $r - k$, so that X is affine by (3.7). $\square$

3.3. Let A be a noetherian ring, and let $\mathfrak{a}$ be an ideal of X .

(a) Show that $\Gamma_{\mathfrak{a}}(\cdot)$ (II, Ex. 5.6) is a left-exact functor from the category of A -modules to itself. We denote its right derived functors, calculated in $\text{Mod}(A)$, by $H_{\mathfrak{a}}^i(\cdot)$.

(b) Now let $X = \text{Spec } A$, $Y = V(\mathfrak{a})$. Show that for any A -module M ,

$$H_{\mathfrak{a}}^i(M) = H_Y^i(X, \widetilde{M}).$$

(c) For any i , show that $\Gamma_{\mathfrak{a}}(H_{\mathfrak{a}}^i(M)) = H_{\mathfrak{a}}^i(M)$.

Proof. (a) Let $0 \rightarrow M' \xrightarrow{\phi} M \xrightarrow{\psi} M''$ be a left exact sequence in Mod_A , and consider the following diagram.

$$\begin{array}{ccccccc} 0 & \longrightarrow & M' & \xrightarrow{\phi} & M & \xrightarrow{\psi} & M'' \\ & & \uparrow & & \uparrow & & \uparrow \\ 0 & \longrightarrow & \Gamma_{\mathfrak{a}}(M') & \xrightarrow{\phi_{\mathfrak{a}}} & \Gamma_{\mathfrak{a}}(M) & \xrightarrow{\psi_{\mathfrak{a}}} & \Gamma_{\mathfrak{a}}(M'') \end{array}$$

We clearly have $\phi_{\mathfrak{a}}$ injective and $\text{im } \phi_{\mathfrak{a}} \subset \ker \psi_{\mathfrak{a}}$. It suffices to show $\ker \psi_{\mathfrak{a}} \subset \text{im } \phi_{\mathfrak{a}}$.

Take $m \in \ker \psi_{\mathfrak{a}} \subset \ker \psi$, and $m' \in M'$ such that $\phi(m') = m$. We have $n > 0$ such that $0 = \mathfrak{a}^n \phi(m') = \phi(\mathfrak{a}^n m')$, whence $\mathfrak{a}^n m' = 0$ by injectivity of ϕ . We conclude that $m' \in \Gamma_{\mathfrak{a}}(M')$, so that $m = \phi_{\mathfrak{a}}(m') \in \text{im } \phi_{\mathfrak{a}}$ gives us exactness.

(b) Let $0 \rightarrow M \rightarrow I^\bullet$ be an injective resolution of M in Mod_A . Then $\widetilde{I}^\bullet$ is a flasque resolution of $\widetilde{M}$ by (3.4). Thus, it suffices to show that $\Gamma_{\mathfrak{a}}(I^\bullet) \cong \Gamma_Y(\widetilde{I}^\bullet)$. This follows from the more general fact that $\Gamma_{\mathfrak{a}}(N) \cong \Gamma_Y(\widetilde{N})$, which is a corollary of (II, Ex. 5.6c).

(c) Let $0 \rightarrow M \rightarrow I^\bullet$ be an injective resolution of M , so that $H_a^\bullet(M) = h^\bullet(\Gamma_a(I^\bullet))$. Then, with boundary maps $d^p : \Gamma_a(I^p) \rightarrow \Gamma_a(I^{p+1})$,

$$\begin{aligned} \Gamma_a(h^p(\Gamma_a(I^\bullet))) &= \Gamma_a\left(\frac{\ker d^p}{\operatorname{im} d^{p-1}}\right) \\ &\supset \frac{\Gamma_a(\ker d^p)}{\operatorname{im} d^{p-1}} \\ &= h^p(\Gamma_a(I^\bullet)), \end{aligned}$$

since $\ker d^p \subset \Gamma_a(I^p)$. That is,

$$\Gamma_a(H_a^\bullet(M)) = H_a^\bullet(M),$$

as desired. $\square$

3.7 Let A be a noetherian ring, let $X = \operatorname{Spec} A$, let $\mathfrak{a} \trianglelefteq A$ be an ideal, and let $U \subset X$ be the open set $X - V(\mathfrak{a})$.

(a) For any A -module M , establish the following formula of Deligne:

$$\Gamma(U, \widetilde{M}) \cong \varinjlim_n \operatorname{Hom}_A(\mathfrak{a}^n, M).$$

(b) Apply this in the case of an injective A -module I to give another proof of (3.4).

Proof. (a) Fix $f \in \mathfrak{a}$, and consider the direct system $\{\operatorname{Hom}_A(\mathfrak{a}^n, M)\}_{n>0}$ with compatibility given by restriction. For any $n > 0$, we have a natural map

$$\begin{aligned} \Psi_{n,f} : \operatorname{Hom}_A(\mathfrak{a}^n, M) &\longrightarrow M_f \\ \phi &\longmapsto \frac{\phi(f^n)}{f^n}. \end{aligned}$$

We see that

$$\begin{aligned} \Psi_{n+1,f}(\phi|_{\mathfrak{a}^{n+1}}) &= \frac{\phi|_{\mathfrak{a}^{n+1}}(f^{n+1})}{f^{n+1}} \\ &= \frac{\phi(f^n)}{f^n} \\ &= \Psi_{n,f}(\phi), \end{aligned}$$

so the $\Psi_{n,f}$ induce a natural map

$$\Psi_f : \varinjlim_n \operatorname{Hom}_A(\mathfrak{a}^n, M) \rightarrow M_f.$$

Surjectivity is clear, as $\Psi_f(f^n \mapsto m) = m/f^n \in M_f$ for arbitrary $m \in M$ and $n > 0$.

Recall that

$$\Gamma(U, \widetilde{M}) = \varprojlim_{f \in \mathfrak{a}} \Gamma(D_f, \widetilde{M}) = \varprojlim_{f \in \mathfrak{a}} M_f,$$

where the inverse system is given by $g \leq f$ iff $\sqrt{(g)} \subset \sqrt{(f)}$, with maps

$$\begin{aligned} \rho_g^f : M_f &\longrightarrow M_g \\ \frac{m}{f^n} &\longrightarrow \frac{mu^n}{g^{kn}}, \end{aligned}$$

where $u \in A$ such that $g^k = uf$ is supplied by $(g) \subset \sqrt{(g)} \subset \sqrt{(f)}$. We compute

$$\begin{aligned} \rho_g^f \Psi_f(\phi) &= \rho_g^f \left(\frac{\phi(f^n)}{f^n} \right) \\ &= \frac{\phi(f^n)u^n}{g^{kn}} \\ &= \frac{\phi(f^n u^n)}{g^{kn}} \\ &= \frac{\phi(g^{kn})}{g^{kn}} \\ &= \Psi_g(\phi), \end{aligned}$$

so that the Ψ_f glue to give us a natural morphism $\Psi : \varinjlim \text{Hom}_A(\mathfrak{a}^n, M) \rightarrow \Gamma(U, \widetilde{M})$. It suffices to show that this map is indeed an isomorphism.

We begin by showing the map is injective. If $\Psi(\phi) = 0$, then $\Psi(\phi)|_{D_f} = 0$ for all $f \in \mathfrak{a}$. Thus, for any $f \in \mathfrak{a}$ there exists $n > 0$ such that $\phi(f^n)/f^n = 0 \in M_f$. That is,

$$\phi(f^{n+k}) = f^k \phi(f^n) = 0$$

for $k \geq 0$. Taking generators $f_1, \dots, f_r \in \mathfrak{a}$ supplied by noetherianity of A , we have that

$$\phi(\mathfrak{a}^n) = \phi((f_1, \dots, f_r)^n) = 0$$

for $n \gg 0$. Hence, $\phi = 0 \in \varinjlim \text{Hom}_A(\mathfrak{a}^n, M)$.

We conclude by showing that Ψ is surjective. For any $s \in \Gamma(U, \widetilde{M})$, noetherianity gives a finite cover $\{D_{f_i}\}_{i=1}^r$ of U , where $\mathfrak{a} = (f_1, \dots, f_r)$. It suffices to find $\phi \in \varinjlim \text{Hom}_A(\mathfrak{a}^n, M)$ such that $\Psi_{f_i}(\phi) = s_i := s|_{D_{f_i}}$ for each $i = 1, \dots, r$. Recall that Ψ_{f_i} is surjective, so we can choose ϕ_i such that $\Psi_{f_i}(\phi_i) = s_i$. Then $s_i = \phi_i(f_i^n)/f_i^n$ for $n \gg 0$, so

$$\frac{\phi_i(f_i^n)f_j^n}{f_i^n f_j^n} = s_i|_{D_{f_j}} = s_j|_{D_{f_i}} = \frac{f_i^n \phi_j(f_j^n)}{f_i^n f_j^n}$$

for i, j . Thus,

$$\phi_i(f_i^n f_j^n) = \phi_j(f_i^n f_j^n) \in M$$

for $n \gg 0$.

We define a map $\phi : \mathfrak{a}^n \rightarrow M$ for $n \gg 0$ on the generators $\{f_1^{e_1} \cdots f_r^{e_r}\}_{\sum_i e_i = n}$ of $\mathfrak{a}^n = (f_1, \dots, f_r)^n$ by

$$\phi(f_1^{e_1} \cdots f_r^{e_r}) = \phi_i(f_1^{e_1} \cdots f_r^{e_r}),$$

where i maximizes e_i . If $e_i = e_j$ for some i, j , then $e_i, e_j \geq n/r \gg 0$, so that

$$\phi_i(f_1^{e_1} \cdots f_r^{e_r}) = \phi_j(f_1^{e_1} \cdots f_r^{e_r})$$

by the above. If we can write a generator as $f_1^{e_1} \cdots f_r^{e_r} = a f_1^{l_1} \cdots f_r^{l_r}$, where $e_i \neq l_j$ and $a \in A$, then we can remove the former term from our generating set. It suffices, then, to show that ϕ is an A -module homomorphism.

Fix some $n > 0$ so that there are well-defined representatives $\phi_i : \mathfrak{a}^n \rightarrow M$ for each $i = 1, \dots, r$. Consider the ascending chain $\{\ker(\phi_i - \phi_j)|_{\mathfrak{a}^k}\}_{k \geq n}$, which stabilizes by noetherianity of A . Take k such that $\ker(\phi_i - \phi_j)^k = \ker(\phi_i - \phi_j)^{k+1}$.

(b) Fix an injective A -module I , let $V \subset U$ be another open set, and let $\mathfrak{b} \trianglelefteq A$ an ideal such that $V = X - V(\mathfrak{b})$. Then $V(\mathfrak{a}) \subset V(\mathfrak{b})$, so $\sqrt{\mathfrak{a}} \supset \sqrt{\mathfrak{b}}$. A noetherian implies $\mathfrak{a}, \mathfrak{b}$

finitely generated, so we can find $m > 0$ such that $\mathfrak{a} = \sqrt{\mathfrak{a}}^m$ and $\mathfrak{b} = \sqrt{\mathfrak{b}}^m$. Then $\mathfrak{a} \supset \mathfrak{b}$, so I injective implies the restriction map

$$\mathrm{Hom}_A(\mathfrak{a}^n, I) \rightarrow \mathrm{Hom}_A(\mathfrak{b}^n, I)$$

is surjective for any n . A direct limit of epimorphisms is itself an epimorphism, so

$$\Gamma(U, \tilde{I}) \rightarrow \Gamma(V, \tilde{I})$$

is surjective by (a). □

- 4.1. Let $f : X \rightarrow Y$ be an affine morphism of noetherian separated schemes (II, Ex. 5.17). Show that for any quasi-coherent sheaf $\mathcal{F}$ on X , there are natural isomorphisms for all $i \geq 0$

$$H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F}).$$

[Hint: Use (II, 5.8).]

Proof. (II, 5.8) gives us that $f_*\mathcal{F}$ is quasi-compact, since X is noetherian. Thus, it suffices by (4.4) and X, Y noetherian separated to give an isomorphism

$$\check{H}^\bullet(\mathcal{U}, \mathcal{F}) \cong \check{H}^\bullet(\mathcal{V}, f_*\mathcal{F})$$

for some open affine covers $\mathcal{U}$ of X and $\mathcal{V}$ of Y .

Take $\mathcal{V} = \{V_i\}$ to be some open affine cover of Y . f affine gives us that each $f^{-1}(V_i)$ is affine, so we have an open affine cover $\mathcal{U} = \{f^{-1}(V_i)\}$ of X . Noting that

$$\begin{aligned} C^p(\mathcal{V}, f_*\mathcal{F}) &= \prod_{i_0 < \dots < i_p} f_*\mathcal{F}(V_{i_0, \dots, i_p}) \\ &= \prod_{i_0 < \dots < i_p} \mathcal{F}(f^{-1}(V_{i_0, \dots, i_p})) \\ &= \prod_{i_0 < \dots < i_p} \mathcal{F}(U_{i_0, \dots, i_p}) \\ &= C^p(\mathcal{U}, \mathcal{F}) \end{aligned}$$

gives us a natural isomorphism of chain complexes $C^\bullet(\mathcal{U}, \mathcal{F}) \cong C^\bullet(\mathcal{V}, f_*\mathcal{F})$, so in particular an isomorphism $\check{H}^\bullet(\mathcal{U}, \mathcal{F}) \cong \check{H}^\bullet(\mathcal{V}, f_*\mathcal{F})$ of homology. This completes the proof. □

- 4.3. Let $X = \mathbb{A}_k^2 = \mathrm{Spec} k[x, y]$, and let $U = X - \{(0, 0)\}$. Using a suitable cover of U by open affine subsets, show that $H^1(U, \mathcal{O}_U)$ is isomorphic to the k -vector space spanned by $\{x^i y^j : i, j < 0\}$. In particular, it is infinite dimensional. (Using (3.5), this provides another proof that U is not affine.)

Proof. Consider the cover $\mathcal{V} = \{V_x, V_y\}$ of U , where $V_x = D(x)$ and $V_y = D(y)$. Open immersions are separated, so X separated implies U separated. Noetherianity of X gives us that U is locally noetherian. $U = V_x \cup V_y$ is a union of finitely many affine—in particular quasi-compact—open subschemes. Thus, U is itself quasi-compact, and hence noetherian. (4.4) gives us

$$H^\bullet(U, \mathcal{O}_U) \cong \check{H}^\bullet(\mathcal{V}, \mathcal{O}_U).$$

We write

$$\begin{aligned} C^\bullet(\mathcal{V}, \mathcal{O}_U) &= [0 \rightarrow \mathcal{O}_U(V_x) \oplus \mathcal{O}_U(V_y) \xrightarrow{d} \mathcal{O}_U(V_x \cap V_y) \rightarrow 0] \\ &= [0 \rightarrow k[x^{\pm 1}, y] \oplus k[x, y^{\pm 1}] \xrightarrow{d} k[x^{\pm 1}, y^{\pm 1}] \rightarrow 0], \end{aligned}$$

where $d(f(x, y), g(x, y)) = f(x, y) - g(x, y)$. As a vector space, then, $\text{im } d$ is spanned by $\{x^i y^j : i \geq 0\} \cup \{x^i y^j : j \geq 0\}$. Then

$$\begin{aligned} H^1(U, \mathcal{O}_U) &= \text{cok } d \\ &= \frac{k[x^{\pm 1}, y^{\pm 1}]}{\text{im } d} \\ &\cong k\langle x^i y^j : i, j < 0 \rangle, \end{aligned}$$

where $k\langle X \rangle$ denotes the k -span of X . □

4.7. Let X be a subscheme of $\mathbb{P}_k^2$ defined by a single homogeneous equation $f(x_0, x_1, x_2) = 0$ of degree d . (Do not assume f is irreducible.) Assume that $(1, 0, 0) \notin X$. Then show that X can be covered by the two open affine subsets $U = X \cap \{x_1 \neq 0\}$ and $V = X \cap \{x_2 \neq 0\}$. Now, calculate the Čech complex explicitly, and thus show that

$$\begin{aligned} \dim H^0(X, \mathcal{O}_X) &= 1, \\ \dim H^1(X, \mathcal{O}_X) &= \frac{1}{2}(d-1)(d-2). \end{aligned}$$

Proof. Since $(1, 0, 0) \notin X$, we have $X = U \cup V$. Further,

$$\begin{aligned} U &= \text{Spec} \left(\frac{k[\frac{x_0}{x_1}, \frac{x_2}{x_1}]}{(f(\frac{x_0}{x_1}, 1, \frac{x_2}{x_1}))} \right), \\ V &= \text{Spec} \left(\frac{k[\frac{x_0}{x_2}, \frac{x_1}{x_2}]}{(f(\frac{x_0}{x_2}, \frac{x_1}{x_2}, 1))} \right) \end{aligned}$$

are affine. We compute

$$C^\bullet = [0 \rightarrow \frac{k[\frac{x_0}{x_1}, \frac{x_2}{x_1}]}{(f(\frac{x_0}{x_1}, 1, \frac{x_2}{x_1}))} \oplus \frac{k[\frac{x_0}{x_2}, \frac{x_1}{x_2}]}{(f(\frac{x_0}{x_2}, \frac{x_1}{x_2}, 1))} \xrightarrow{d} \frac{k[y_1, y_2^{\pm 1}]}{f}]$$

□